The steeper 1/f slope of MEG power spectrum is associated with stronger gamma synchrony, measured via inter-trial phase coherence (ITPC) in the 40-Hz auditory steady-state response

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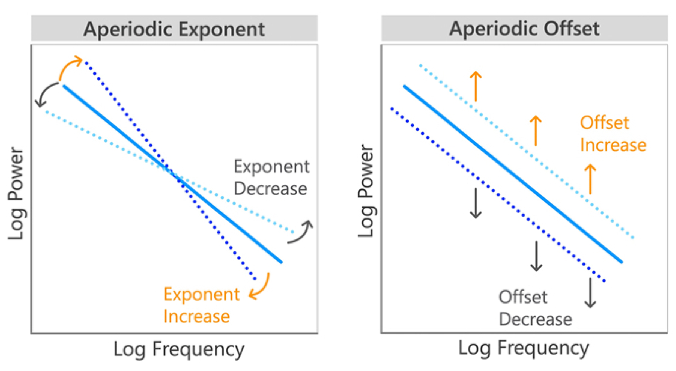
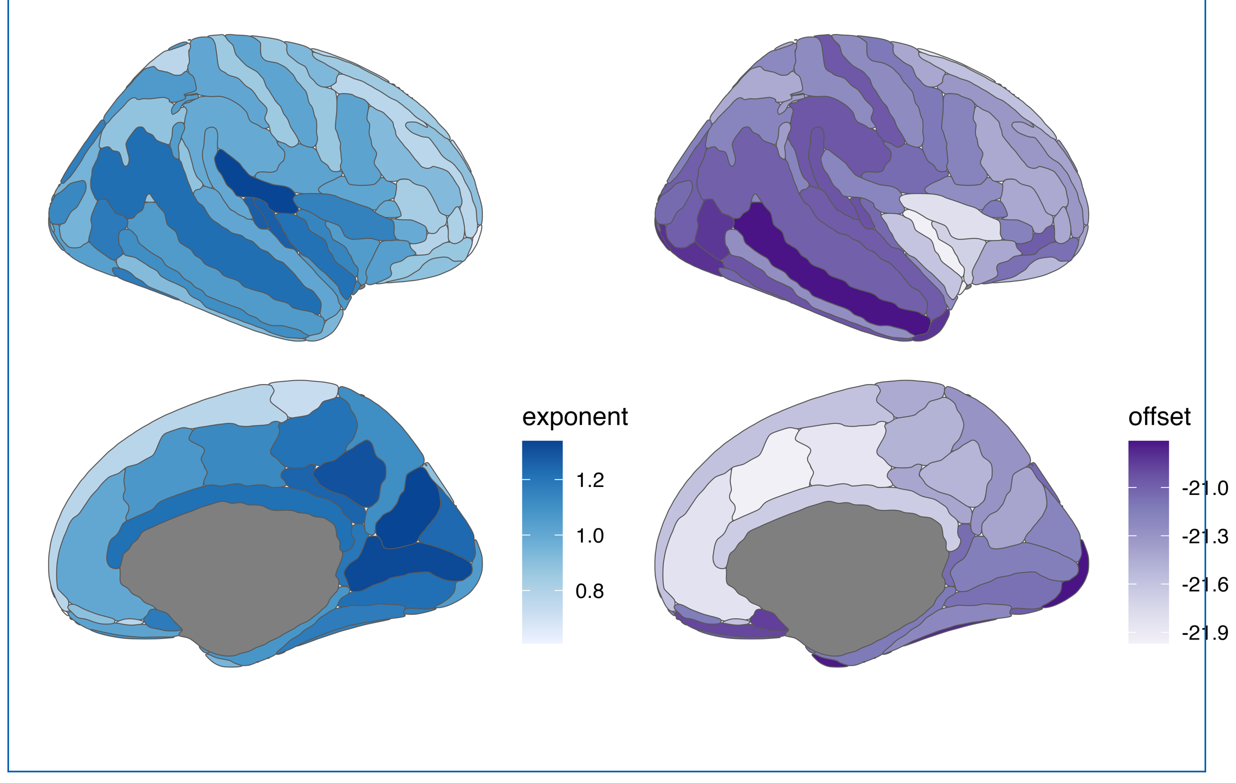
Hypothesis. We explored whether a steeper slope of the 1/f PSD component (i.e., a larger spectral exponent) in MEG signals is associated with higher auditory steady-state response (ASSR) synchrony, as measured by inter-trial phase coherence (ITPC) during 40 Hz auditory stimulation.

Introduction

- Power spectral density (PSD) of electrophysiological activity may be parametrized by oscillatory components and aperiodic component, where the power of the signal decay with a 1/f power law
- The timescale and magnitude of this aperiodic activity can be described by the slope and offset of this 1/f decay [2]
- Simplified formula describing 1/f component of the PSD of the signal: $L(f) = b - \log(k + f^{\chi})$; The exponent χ is related to the negative slope of the power spectrum in log-log space, that is, the inverse relationship between power and frequency; The offset b describe the uniform shift in power across frequencies
- These parameters are most likely governed by waxing and waning of oscillatory components, and can be associated to biological mechanisms (e.g. excitation:inhibition balance as summed by [7], cognitive processes, and group differences).
- Graphical relationship of spectral power and offset and exponent (from [3])

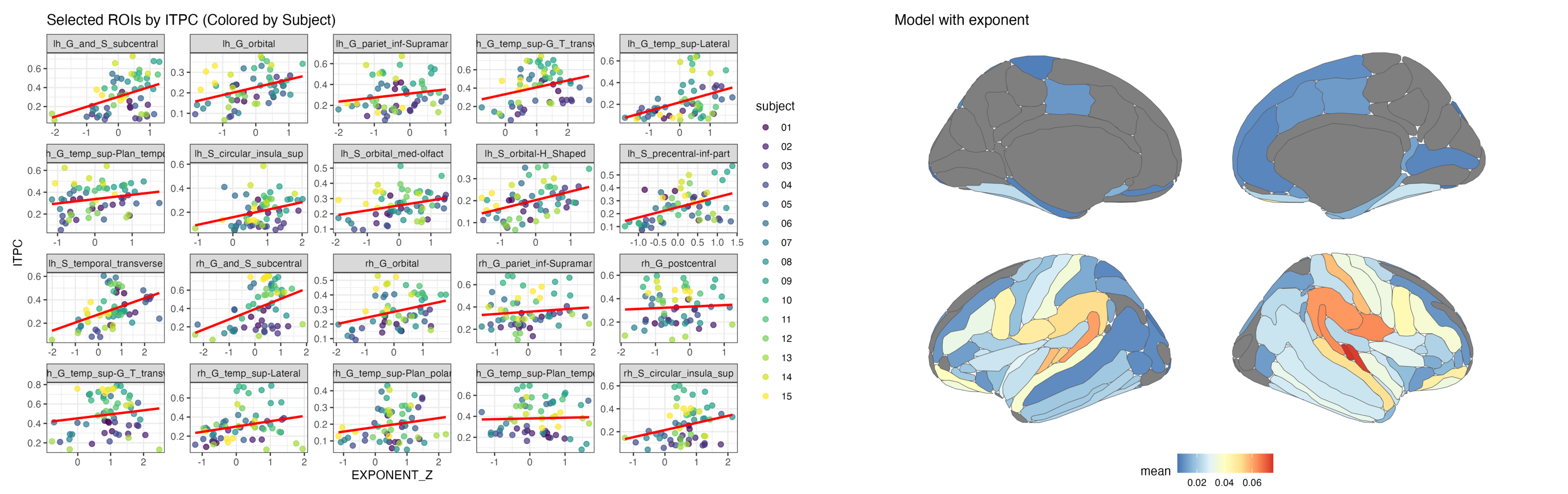


1 Spectral Parametrization. Mean values of estimated exponent and offset describing the 1/f part of spectral power across different regions (right hemisphere).



- Synchronization of brain activity in the gamma frequency is a hallmark of brain function, and an indicator of malfunctioning[5].
- Auditory stimulation in the gamma range is a reliable way to experimentally investigate this phenomenon, looking at auditory steady-state responses (ASSR), the effect of entrainment of brain activity to the frequency and phase of external auditory stimuli.
- ASSR are usually investigated by looking at the inter-trial phase coherence (ITPC) [5].

2 Relationship between the slope (parametrized by exponent and the estimated ITPC). Figure on the left: Relationship between Spectral Exponent (z-scored) and ITPC across selected ROIs. Red line indicates linear fit. Data points colored by subject.



Data

- We re-analysed MEG data from n=15 healthy participants, stimulated with a sound wave modulated at 40 Hz, presented in 180 trials for one second, with the onset of two consecutive trials separated by at least 2 s [5].
- The data were preprocessed, projected into Destrieux brain atlas

Methods

- The PSD of the spectrum were computed for 1s second pre-stimulus period and stimulation period for each trial (and averaged later) and for each region. The slope and exponent were computed using specparam toolbox [2].
- ITPC was computed, the value averaged in the intervals $[-500, -200]$ ms for prestimuli and $[400, 700]$ ms after sound onset for each of the regions using the wavelet approach described in [5].
- The influence of the stimulus and the exponent on the ITPC were tested by modelling the different relationships with an integrative Bayesian Multilevel Model [1]
- This methods allows for better integration of the spatial information and control of multiplicity avoding information loss of univariate approach followed by the correction for multiple testing.
- Effect estimation and region-level inferences are based on posterior distributions of group differences.

Following models were tested:

1 ITPC ∼ S (Stimulus)

2 ITPC ∼ Exponent + Stimulus

3 Exponent ∼ Stimulus

5 Offset ∼ Stimulus

- For investigation of the the relationship we fit the model

$$\mathrm{ITPC_{stim}} \sim 1 + \mathrm{exponent}_z + (1 + \mathrm{exponent}_z \mid \mathrm{subject}) + (1 \mid \mathrm{ROI})$$

with z-scored Exponent. The model was fitted using the `brms` package and [1].

Results (Box 2)

Spatial Distribution & Stability:

- Exponent and offset varied systematically across brain regions (Box 1)
- Exponent remained stable during auditory stimulation (no evidence for Stimulus effect)
- Offset showed no systematic changes across conditions

ITPC Response to Auditory Stimulation:

- Strong stimulus effect on ITPC (baseline vs. stimulation) in auditory cortex, temporal lobe, inferior sensorimotor regions, and inferior frontal regions
- Right hemisphere showed somewhat stronger effects

Exponent-ITPC Relationship:

- Positive relationship between exponent and ITPC during stimulation
- Steeper spectral slopes associated with stronger phase-locking
- Regional specificity: Strongest effects in right superior temporal gyrus, temporal pole, insula, and inferior frontal regions—overlapping with regions showing robust ASSR responses
- Offset showed no significant relationship with ITPC

Statistical Approach:

- Bayesian multilevel modeling accounts for spatial structure and multiple comparisons through hierarchical pooling
- Posterior distributions and P^{+} quantify uncertainty in estimates

Conclusion

- The differences observed might simply reflect increased "noise" in the time-domain signal, potentially diminishing SNR and synchrony. Alternatively, from a biological perspective, this effect might be linked with excitation-inhibition balance, which influence both synchrony measures and spectral slope.
- The effects of the aperiodic slope on the estimation of the instantaneous phase and the ITPC [6] could be considered in further analysis as well as a effect of higher gamma power for aperiodic slope estimation.

Acknowledgments: For visualising the results ggseg library was used [4]

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